

Gregory Burd

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Overview

Senior Staff Software Engineer with 30+ years experience in production environments from embedded to cloud-scale at all layers of the stack with a strong focus on data storage, consistency, and networking. Depth of knowledge in writing code and operating and administering systems in production. Comfortable in many roles, including individual contributor, architecture, technical lead, and product definition/management.

Professional Skills

- In-depth knowledge of operating systems, networking, and data storage
- Practical experience with distributed systems design, deployment, and maintenance at scale
- Experienced programmer in C, Scala, Java, Erlang, Elixir, Python
- Depth of knowledge in UNIX systems including Linux, BSD, Solaris, macOS (OS/X, NeXTSTEP)
- Excellent problem-solving skills, innovative yet well-reasoned solutions architect
- Strong communication skills including the production of technical materials
- Extensive knowledge of software engineering principles, tools, and languages
- Seasoned mentor, team member, and individual contributor

Employment

Klar, Director of Engineering – Cambridge, MA – July 2022 - current

Reduced database infrastructure costs by 60% and aligned costs with customer demand. Reduced database incidents from weekly to infrequent through education and hands-on changes to the 100 microservices. Introduced a new functions-style method for delivering on data engineering tasks reducing complexity for engineers better focused on analysis.

BlockFi, Senior Staff Software Engineer – Cambridge, MA – June 2020 - June 2022

Designed and delivered observability stack at BlockFi (logs, metrics, tracing, alerts), grew the team to 8 then handed it off to a team member I groomed and promoted. I joined the data storage team and oversaw the health of Postgres as well as other data engineering issues.

Humio, Principal Software Engineer – Cambridge, MA – September 2018 - April 2020

I worked on the backend team to create an innovative index-free filter-based storage system able to find needles spread across multi-petabyte archives of log and metric data by intelligently filtering data to reduce IO to the absolute minimum and then exploit modern CPU/memory architecture despite being on the JVM (Scala).

Onshape, Principal Software Engineer, Architect – Cambridge, MA – January 2017 - October 2018

I worked on data storage responsible for primary, object, and long-term storage as well as caching. I was responsible for taking Onshape from the pre-launch/1.0 data architecture based entirely on MongoDB and Memcached to something that will provide more consistent performance at scale. To that end, I authored a new client layer that used Cassandra while providing causal consistency and in-application caching and nesting transactions for accessing model objects via a friendly Java-based DSL (<https://github.com/gburd/helenus> and <https://github.com/gburd/memcached-jcache>).

Akamai Technologies, Senior Software Development Engineer – Cambridge, MA – July 2015 - December 2016

I was a member of the Labs group at Akamai. I owned the database component (design, implementation, deployment, automation, and ongoing management) for a WebRTC-based peer and multi-party video conferencing platform. In addition, I customized a Linux distro to provide an always-on, low-maintenance kiosk-style video-wall experience combining an Intel NUC and large television and deployed this into Akamai's lobby as well as other locations within the company.

Amazon AWS, Senior Software Development Engineer – Cambridge, MA – July 2014 - July 2015

I wrote and deployed software for the Storage Gateway service's data and control plane as well as for the virtualized on-premise iSCSI gateway. I orchestrated the zero-downtime upgrade of thousands of customers remotely to mitigate the Bash ShellShock bug. I investigated and produced a design to incorporate de-duplication techniques to lower the overhead of storage and speed up the time to perform backups of customer data into the service's data plane. I represented our service in weekly operations meetings and performed root cause analysis for many system failures.

Basho Technologies, Software Architect and Product Manager – Cambridge, MA – Jul 2011 - Apr 2014

I authored node-level data storage and asynchronous thread queues for Riak in Erlang and C. I identified TCP node-level communication using µTP. I built a layer to isolate Erlang's BEAM virtual machine from native code to avoid a scheduler collapse. I investigated alternatives to TCP for intra-node work, specifically reviewing µTP (micro transport protocol, used by BitTorrent). I built a pure-Erlang local LSM-style ordered storage layer called HanoiDB and investigated other local storage alternatives.

Oracle Corporation, Product Manager – Burlington, MA – Feb 2006 - Mar 2011

Sleepycat Software, Software Engineer/Product Manager – Boston, MA – Jul 2002 - Feb 2006

KnowNow, Software Engineer/Technical Alliance Manager – Menlo Park, CA – Dec 2000 - Jun 2002

Primix Solutions, Software Engineer/Managing Consultant – Watertown, MA – Sep 1996 - Jun 1999

Sun Microsystems, Software Engineer Member of Technical Staff (MTS-4) – Bay Area, CA – Jul 1999 - Nov 2000

Marble Associates, Software Engineer/Consultant – San Jose, CA then later Watertown, MA – August 1993 - August 1996

NeXT Computer, Developer Support Team – Redwood City, CA – May 1992 - September 1992